

SCHOOL OF ELECTRICAL AND INFORMATION ENGINEERING  
**ELEC2302/9302 Signals and Systems — Lab 2**  
Fourier analysis

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## Equipment

- Telecommunication Instructional Modelling System (TIMS) with the following modules:

| Module                        | Qty | Type           |
|-------------------------------|-----|----------------|
| Multiplier                    | 1   | Movable module |
| Tuneable Low-pass Filter      | 1   | Movable module |
| Utilities                     | 1   | Movable module |
| Voltage-controlled Oscillator | 1   | Movable module |
| Frequency Counter             | 1   | Fixed module   |
| PC-based instrument inputs    | 1   | Fixed module   |
| Variable DC                   | 1   | Fixed module   |
| Master Signals                | 1   | Fixed module   |

- Banana-plug cables, for connections between TIMS modules.
- A PC with the Signals and Systems (S&S) software and PicoScope 7 installed.

## Lab information

- Bring an exercise book: the observations you record in it are the record of the lab.
- There are three checkpoints in this sheet, each worth 1% of the unit mark. At each one, show your work to a tutor and answer the question they ask.

## 1 Lab Theory

The goal of this lab is to learn to measure the spectrum of a periodic signal experimentally.

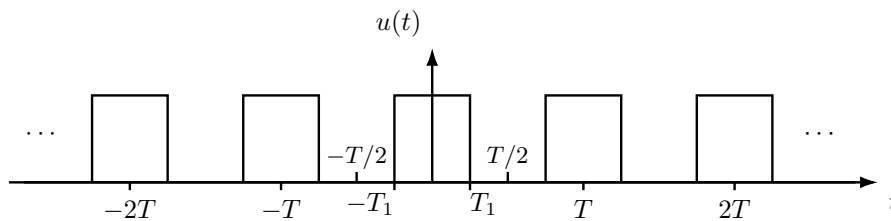
### Fourier coefficients of a square wave

Lets look at the periodic square wave  $u(t)$  sketched in Figure 1 which has period  $T$  and is defined over one period by

$$u(t) = \begin{cases} 1, & |t| < T_1, \\ 0, & T_1 < |t| < T/2. \end{cases}$$

Equivalently,  $u(t)$  is the pulse  $p_{2T_1}(t)$  repeated with period  $T$ ,

$$u(t) = \sum_{k=-\infty}^{\infty} p_{2T_1}(t - kT).$$



**Figure 1:** The periodic square wave  $u(t)$ , drawn for  $T = 4T_1$

As this is a periodic signal, we can calculate its Fourier series. A general Fourier series is given by

$$u(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}, \quad c_k = \frac{1}{T} \int_{-T/2}^{T/2} u(t) e^{-jk\omega_0 t} dt, \quad \omega_0 = \frac{2\pi}{T}.$$

For this particular square wave signal, the Fourier coefficients can be found to be

$$c_0 = \frac{2T_1}{T}, \quad c_{k>0} = \frac{1}{\pi k} \sin\left(\frac{2\pi k T_1}{T}\right)$$

### Product of two sinusoids

We can find using Euler's identity that the product of two sinusoids of frequencies  $\omega_1$  and  $\omega_2$  is a sum of two sinusoids given by

$$A \cos(\omega_1 t) \cdot B \cos(\omega_2 t) = \frac{1}{2} AB [\cos((\omega_1 + \omega_2)t) + \cos((\omega_1 - \omega_2)t)].$$

### Lowpass filtering

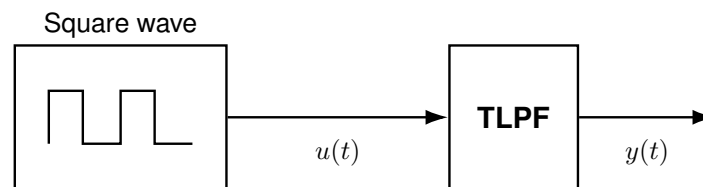
A low-pass filter filters out high frequencies, allowing only low frequencies to pass. An ideal low-pass filter has a gain of 1 below its cutoff frequency, and a gain of 0 for frequencies above.

In practice, filters have smooth frequency responses so we define the cutoff frequency to be  $f_{-3dB}$ , where the gain becomes  $-3dB$ , which corresponds to approximately 50% reduction in power.

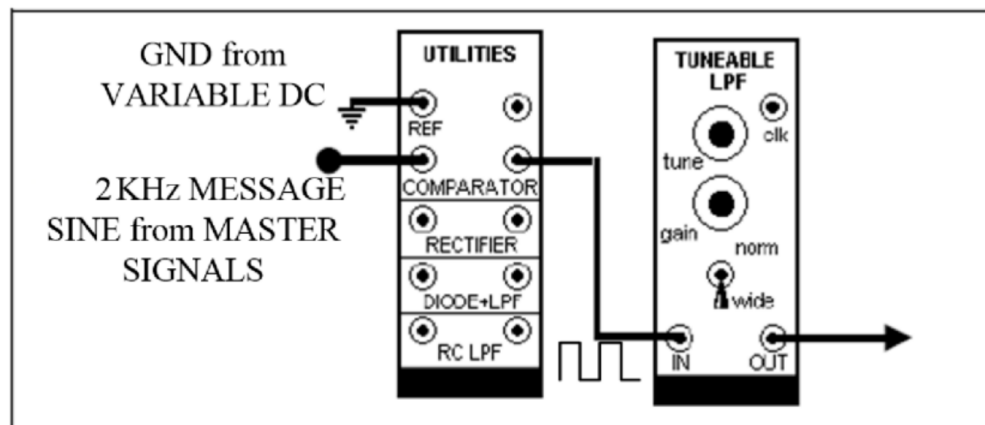
## 2 Experiments

### 2.1 Smoothing of a square wave

We first investigate what happens to a square wave when it is passed through a low-pass filter. The block diagram is shown in Figure 2 and the TIMS model in Figure 3.



**Figure 2:** Block diagram for smoothing of a square wave



**Figure 3:** TIMS model of Figure 2

#### 2.1.1 Procedure

1. Connect the input to the **Tuneable Low-pass Filter** to CH-A, and the output to CH-B of the **PC-based instrument inputs**
2. Set the **Tuneable Low-pass Filter** range to WIDE and its TUNE knob fully clockwise. Put the gain knob marker at 12 o'clock
3. Set up the **Utilities** module to output a square wave by connecting its REF to GND from **Variable DC** and the 2KHz sine from **Master Signals**
4. Connect the **Utilities** comparator output to the input of the **Tuneable Low-pass Filter**. Confirm that your connections look like what is shown in Figure 3.

With this done, **Utilities** should be producing a square wave at approximately 2.083 kHz which you can see on CH-A!

### 2.1.2 Tasks

- Display the TLPF input  $u(t)$  and output  $y(t)$  and sketch the shapes in your notebook. Measure and label  $T$  and  $T_1$  on your sketch for the input square wave.

To measure the TLPF cut-off frequency, connect the red CLK socket to the TTL input of the **Frequency Counter**. Then calculate

$$f_{-3\text{dB}} = \frac{\text{counter reading}}{100}.$$

Slowly turn the TUNE knob counter-clockwise while watching the output. The distorted waveform will gradually become a smooth sinusoid.

- Record the point at which the output becomes a smooth sinusoid. What is  $f_{-3\text{dB}}$  of the TLPF there, and what is the frequency of the sinusoid?
- Switch to the NORMAL range. With the TUNE knob fully clockwise, record the output waveform. Repeat with the TUNE knob fully counter-clockwise.

#### Checkpoint 1

Show a tutor your sketches, with the measurements marked on them. Which terms of the Fourier series survive at each setting of the TUNE knob, and how does that account for the shape of the output?

## 2.2 Fourier series representation of a square wave

Write the product identity from Section 1 with one known (ref) and one unknown frequency (test),

$$A \cos(\omega_{\text{test}} t) \cdot B \cos(\omega_{\text{ref}} t) = \frac{1}{2} AB \left[ \cos((\omega_{\text{test}} + \omega_{\text{ref}})t) + \cos((\omega_{\text{test}} - \omega_{\text{ref}})t) \right].$$

To find an unknown amplitude and frequency, we can multiply a signal by a known reference sinusoid. The product contains sinusoids of two frequencies, called beat signals. A tuned lowpass filter can remove the higher beat and leave

$$\frac{1}{2} AB \cos((\omega_{\text{test}} - \omega_{\text{ref}})t).$$

The amplitude of what is left is proportional to  $A$  as long as the reference amplitude  $B$  is held fixed. In this experiment, we will find the Fourier series for our square wave this way by isolating this beat frequency.

### 2.2.1 Procedure

1. Disconnect the **Utilities** square wave output from the TLPF and connect it to the **Multiplier** X input.
2. Connect the **Voltage-controlled Oscillator** (VCO) SINE output to the **Multiplier** Y input.
3. Connect the VCO  $V_{in}$  input to the DC output from the **Variable DC** module. Set the gain control, which gives fine control of the VCO frequency, to 10 o'clock and the DC control of the associated **Variable DC** to 12 o'clock.
4. Connect the **Multiplier** output to the **Tuneable Low-pass Filter** IN.

5. Set the **Tuneable Low-pass Filter** to the NORMAL range with its tuning control at 9 o'clock. Confirm that your setup matches Figure 5.
6. Connect the VCO output to CH-C of **PC-based instrument inputs** so you can record its frequency and amplitude for use in calculations.

The **Voltage Controlled Oscillator** produces a sine output that can act like our reference signal. We can use the low-pass filter to filter out everything but a lower beat frequency.

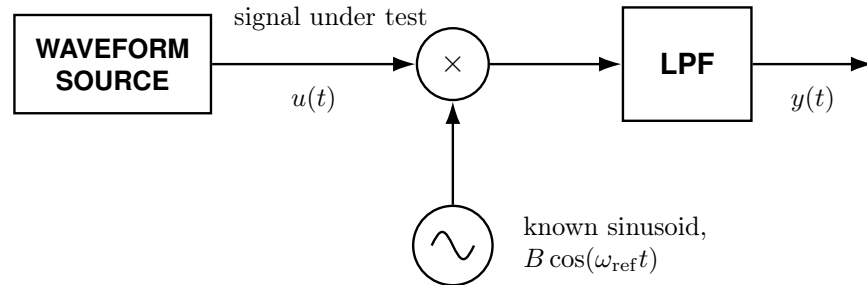


Figure 4: Block diagram of the technique

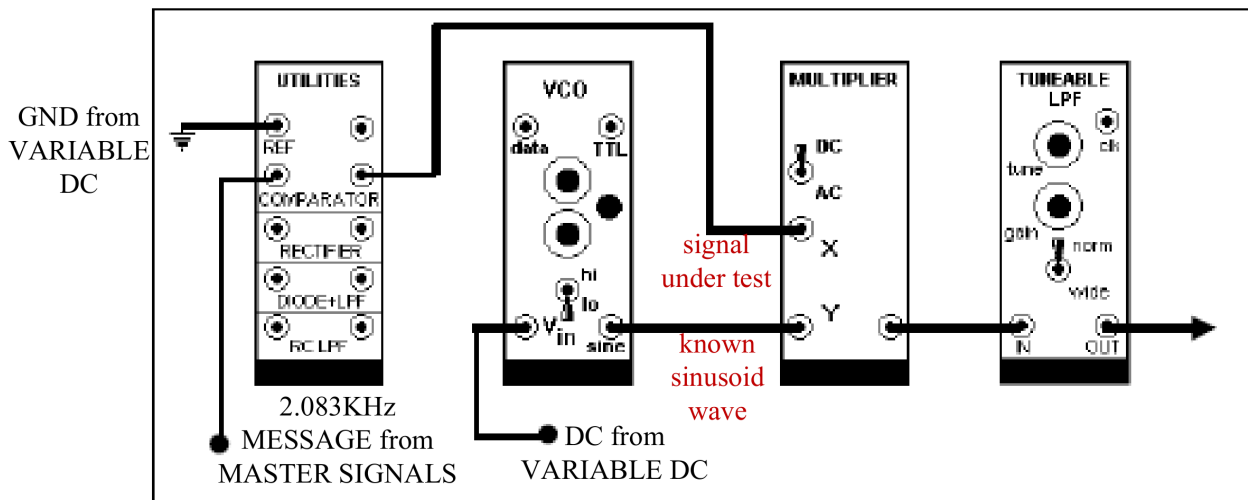


Figure 5: TIMS model of Figure 4

### 2.2.2 Tasks

- Using the frequency control knob  $f_0$ , set the VCO frequency close to 2.083 kHz, but at least 200 Hz below it, and display the beat signal at the output of the TLPF. Record the frequency and the amplitude of the beat signal.
- Derive the frequency and the amplitude of the test sinusoid from the beat signal.

#### Checkpoint 2

Show your tutor the beat signal and your amplitude calculations.

- Repeat near 4.176 kHz, 6.249 kHz and 8.333 kHz, staying at least 200 Hz away in each case. Can you find the target signals there?
- Tabulate your findings in the form below and compare them with your prework calculation.

| Reference from VCO |               | Measured on scope |                | Signal under test (calculated) |                                     |                             |
|--------------------|---------------|-------------------|----------------|--------------------------------|-------------------------------------|-----------------------------|
| VCO frequency      | VCO amplitude | Beat frequency    | Beat amplitude | Target frequency               | Target amplitude (beat amplitude/2) | Normalised target amplitude |
| 1.85 kHz           |               |                   |                |                                |                                     |                             |
| 3.86 kHz           |               |                   |                |                                |                                     |                             |
| 5.99 kHz           |               |                   |                |                                |                                     |                             |
| 8.03 kHz           |               |                   |                |                                |                                     |                             |

- Calculate the amplitudes, and their ratios, for the theoretical square wave using the formulas provided in Section 1.

| Coefficient | Frequency | Amplitude | Normalised amplitude $ c_k/c_1 $ |
|-------------|-----------|-----------|----------------------------------|
| $c_0$       |           |           | N/A                              |
| $c_1$       |           |           |                                  |
| $c_2$       |           |           |                                  |
| $c_3$       |           |           |                                  |
| $c_4$       |           |           |                                  |

### Checkpoint 3

Show your tutor your calculated amplitudes. How do the measured amplitudes compare with the coefficients  $c_k$  you calculated in the prework? Optionally, use a spreadsheet program to plot both side by side.